

Math 164 Challenge Problem

Directional derivatives, gradients, double integrals, optimization

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Problem 1.

What is the area of the domain of the function in the given double integral: $\int_0^3 \int_0^x f(x,y) dy dx$ where $f(x,y) = e^x \cos(x) \sin(x) \tan(x) \sec(x)$?

Solution: Recall that volume is given by $V = \text{Base} \times \text{Height}$. If we set $\text{height} = 1$ then the result of the double integral is simply the area of the domain! So all you have to do is set $f(x,y) = 1$ then solve. (It doesn't matter what $f(x,y)$ is!)

$$\int_0^3 \int_0^x 1 dy dx = \int_0^3 x dx = \frac{3^2}{2} = \frac{9}{2}$$

So the area of the domain of the function in the given double integral is $\frac{9}{2}$.

Problem 2.

You just bought a new house! One of your rooms has an interesting ceiling modelled by the function $f(x,y) = x^2 + y^2$ where the xy plane is your floor. Imagine that the floor is a rectangle and the walls go directly up to the ceiling perpendicular to the floor. The rectangular floor can be modelled by the equation $R = [0, 11] \times [0, 12]$ (i.e. your corners of the room are given by the points $(0,0), (11,0), (0,12), (11,12)$). Assume units are in ft .

- (a) What is the highest point in the room (answer should have units of ft)?

Solution: This is an optimization problem. We first find if there are any critical points.

$$f_x = 2x$$

$$f_y = 2y$$

The only critical points are $(0,0)$. If you simply visualize $f(x,y)$, it is obvious that at $(0,0)$ it is an absolute minimum. We have $f(0,0) = 0$. Next we find the absolute max/min on the boundaries:

- i Along the x-axis: $y = 0$ and $0 \leq x \leq 11$. So $f(x,y) = x^2$ from $0 \leq x \leq 11$. Clearly the min is $f = 0$ and the max is $f = 11^2 = 121$
- ii Along the y-axis: $x = 0$ and $0 \leq y \leq 12$. So $f(x,y) = y^2$ from $0 \leq y \leq 12$. Clearly the min is $f = 0$ and the max is $f = 12^2 = 144$
- iii Along the line $x = 11$, $0 \leq y \leq 12$. So $f(x,y) = 11^2 + y^2$ from $0 \leq y \leq 12$. Clearly the min is $f = 11^2 = 121$ and the max is $f = 11^2 + 12^2 = 265$
- iv Along the line $y = 12$, $0 \leq x \leq 11$. So $f(x,y) = x^2 + 12^2$ from $0 \leq x \leq 11$. Clearly the min is $f = 12^2 = 144$ and the max is $f = 11^2 + 12^2 = 265$

We see that the absolute minimum of the function in our domain occurs at $(0,0)$ with $f(0,0) = 0$. Note that the critical point is the absolute min. The absolute maximum in our domain occurs at $(11,12)$ with $f(11,12) = 265$. This is precisely our answer. So the highest point in the room occurs in the corner $(11,12)$ with height of $265ft$.

(b) You are holding a balloon with an infinitely long string that floats directly above you and touches the ceiling at all times. Suppose you are in your room at the point $(1,1)$. In what direction should you walk such that the balloon moves away from you at the fastest rate?

Solution: You must understand directional derivative and gradient vector for this problem. The direction in which the balloon moves away from you at the fastest rate is in the direction of the gradient vector! In other words, it is the direction in which the ceiling is rising the fastest. So we find the gradient vector given by $\nabla f(x,y) = \langle 2x, 2y \rangle$. At point $(1,1)$ we have $\nabla f(1,1) = \langle 2, 2 \rangle$. So we must walk in the direction of $\langle 2, 2 \rangle$ for the balloon to move away from us at the fastest rate.

(c) From the problem in part (a), how much additional length of string is pulled at the instant when you walk in the specific direction so that the balloon maintains contact with the ceiling at all times.

Solution: The question is asking in other words, how fast is the ceiling increasing when we walk in the direction of fastest increase (that is the gradient)? Observe that we should have a scalar answer (you can apply units if you want). This problem is the definition of the directional derivative. Recall from workshop that the directional derivative in the direction of the gradient is

simply the magnitude of the gradient. We have

$$D_{\nabla f(1,1)} = |\nabla f(1,1)| = |\langle 2, 2 \rangle| = \sqrt{8}$$

So when we are at the point (1,1) in the room and walk in the direction of the (2,2), at that instant our balloon string increases by $\sqrt{8}ft$.

(d) Fix the width of the room (the x-axis), what should the length of the room (y-axis) be such that the volume of your room is $1500ft^3$.

Solution: This is simply an application of double integrals. Let ℓ be the length of the room. Then we have

$$\int_0^{\ell} \int_0^{11} (x^2 + y^2) dx dy = 1500$$

Solving for ℓ we have

$$\begin{aligned} \int_0^{\ell} \left(\frac{1}{3}x^3 + y^2x \right) \Big|_0^{11} dy &= \int_0^{\ell} \left(\frac{1331}{3} + y^2(11) \right) dy = \frac{1331}{3}y + \frac{y^3}{3}(11) \Big|_0^{\ell} = 1500 \\ \frac{1331}{3}\ell + \frac{\ell^3}{3}(11) &= 1500 \\ 1331\ell + 11\ell^3 - 4500 &= 0 \end{aligned}$$

We use a calculator for the last step and have $\ell = 3.128ft$. This is a really small room. This makes sense because the function $f(x, y) = x^2 + y^2$ increases so fast!